

1. Administrative & Study Identification

Full Study Title: A Phase 1/2 Multiple Expansion Cohort Trial of MRTX1719 in Patients with Advanced Solid Tumors with Homozygous MTAP Deletion
Short Title/Acronym: Phase 1 Study of MRTX1719 in Solid Tumors With MTAP Deletion
Protocol Number: CA240-0007 / MIRATI-1719-001
NCT Number: NCT05245500
Sponsor Name: Bristol-Myers Squibb (formerly Mirati Therapeutics)
Investigational Product: MRTX1719 (PRMT5-MTA inhibitor)
Phase of Development: Phase I/II
Medical Monitor: BMS Clinical Trials Contact Center / Lead Investigator

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: In Phase 1, to evaluate the safety and tolerability of MRTX1719 and determine the Maximum Tolerated Dose (MTD) and Recommended Phase 2 Dose (RP2D). In Phase 2, to evaluate the anti-tumor activity via Objective Response Rate (ORR). **Secondary Objectives:** To characterize the pharmacokinetic (PK) parameters (including C_{max} , T_{max} , AUC) and further assess preliminary clinical efficacy through Duration of Response (DOR), Progression-Free Survival (PFS), and Overall Survival (OS).

2.2 Background & Rationale Homozygous deletions in the *MTAP* gene occur in approximately 10% of all cancers, representing a distinct molecular subset with a poor prognosis. *MTAP* deficiency causes an accumulation of its substrate, methylthioadenosine (MTA), which competitively inhibits the binding of S-adenosyl methionine (SAM) to protein arginine methyltransferase 5 (PRMT5). MRTX1719 is an MTA-cooperative PRMT5 inhibitor specifically designed to target the PRMT5-MTA complex, selectively exploiting this vulnerability to induce synthetic lethality in MTAP-deleted cancer cells while sparing normal, healthy cells.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (First-in-Human / Dose Escalation and Expansion) **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A (Parallel assignment to expansion cohorts based on tumor histology)

3.2 Planned Enrollment & Duration Target **\$N\$:** Approximately 580 evaluable patients **Number of Sites:** Multicenter (Multiple global locations, including major US oncology centers) **Estimated Study Start:** 09-Jun-2022 **Estimated Primary Completion:** 10-Dec-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Histologically confirmed diagnosis of a solid tumor malignancy with homozygous deletion of the *MTAP* gene detected in tumor tissue (incorporating advanced, refractory solid tumors e.g., NSCLC, pancreatic, melanoma). 3. Unresectable or metastatic disease. 4. Presence of a tumor lesion amenable to mandatory biopsy for pharmacodynamic evaluation at baseline and on-study. 5. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1.

4.2 Exclusion Criteria 1. Prior treatment with a PRMT5 or MAT2A inhibitor therapy. 2. Active brain metastases or carcinomatous meningitis. 3. History of significant hemoptysis or hemorrhage, or major surgery within 4 weeks of the first dose of study treatment. 4. History of intestinal disease, major gastric surgery, or other conditions likely to alter the absorption of oral study treatment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: MRTX1719 administered at exploring dose levels (Phase 1 escalation) and the established RP2D (Phase 2 expansion). Patient compliance is monitored via an oral dosing diary. **Route of Administration:** Oral (Capsules and Tablets; assessing relative bioavailability). **Duration of Treatment:** Continuous until disease progression, unacceptable toxicity, or patient withdrawal. Follow-up continues until disease progression, withdrawal, or estimated study completion (Dec 2027).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for oncology patients (e.g., anti-emetics, pain management) excluding prohibited agents. **Prohibited:** Concomitant use of other investigational anticancer therapies, prior PRMT5/MAT2A inhibitors, and strictly forbidden strong CYP3A4 modulators affecting PK.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent (Phase 1/2 trial risks and oral dosing diary compliance), Medical History, Tumor biopsy (MTAP status), Baseline Imaging | | **Baseline** | Day 1, Cycle 1 | First Dose, Intensive PK/PD blood

draws, Vital signs | | **Interim** | Every Cycle (21 to 28 days) | Safety Labs, Efficacy Assessment (Restaging imaging every ~8 weeks), AE monitoring, Inter-cycle remote safety checks as clinically indicated | | **End of Study** | +30 Days post-last dose | Final Vital Signs, Exit Interview, IP Accountability, Survival Follow-up |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: - *Phase 1:* Incidence of Dose-Limiting Toxicities (DLTs) and Treatment-Related Adverse Events (TRAEs). - *Phase 2:* Objective Response Rate (ORR). **Measurement Method:** Clinical safety laboratory panels (Phase 1) and standard radiological tumor imaging assessed per RECIST v1.1 (Phase 2), supported by centralized independent radiological review.

7.2 Secondary & Exploratory Endpoints - Duration of Response (DOR). - Progression-Free Survival (PFS). - Overall Survival (OS). - Pharmacokinetics ($SC_{\max}$, $ST_{\max}$, AUC).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection via investigator assessment, routine clinical laboratory evaluations, and patient symptom reporting at every cycle visit. **Serious Adverse Events (SAEs):** Must be reported to the sponsor (BMS) within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Internal Safety Review Committee (SRC) to evaluate DLTs during the Phase 1 dose-escalation phase before clearing subsequent dose cohorts. Continuous safety signal tracking deployed throughout.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy; Safety Set for all patients receiving at least one dose of MRTX1719; PK-evaluable set. **Sample Size Justification:** Approximately $N=580$ dynamically allocated across dose-escalation and Phase 1b/2 disease-specific expansion cohorts to provide sufficient statistical power for ORR and safety profiling. **Handling of Missing Data:** Time-to-event endpoints (PFS, OS) will employ standard Kaplan-Meier estimates and right-censoring for missing/lost-to-follow-up data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. HCP

education protocols include Investigator meetings and Site Initiation Visits detailing PK/PD sampling requirements and DLT criteria. **Monitoring Plan:** Routine onsite and centralized monitoring by the Sponsor (BMS) to ensure protocol adherence and data integrity. **Audit Trail:** Secure Electronic Data Capture (EDC) system with full audit trails, strict eCRF locking procedures, and source data verification.

11. Study Identifiers & Governance

Primary ID: CA240-0007 (legacy MIRATI-1719-001) **Registry Number:** NCT05245500 **Sponsor/Lead Organization:** Bristol-Myers Squibb **Study Title:** Phase 1 Study of MRTX1719 in Solid Tumors With MTAP Deletion **Official Regulatory Title:** A Phase 1/2 Multiple Expansion Cohort Trial of MRTX1719 in Patients with Advanced Solid Tumors with Homozygous MTAP Deletion

12. Clinical Framework (Oncology Specific)

Primary Condition: Advanced Solid Tumors **Diagnosis Threshold:** Homozygous deletion of the *MTAP* gene confirmed via tumor tissue testing **Medical Methodology:** Targeted PRMT5-MTA synthetic lethality inhibition **Optimization Target:** Maximum Tolerated Dose (MTD) and Recommended Phase 2 Dose (RP2D)

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for advanced, refractory solid tumors (e.g., NSCLC, pancreatic, melanoma) **Education Protocol (HCP):** Investigator meetings and Site Initiation Visits detailing the PK/PD sampling requirements and DLT criteria. **Education Protocol (Patient):** Informed consent documentation regarding Phase 1/2 trial risks and oral dosing diary compliance. **Quality Audit Mechanism:** Centralized independent radiological review of tumor scans (RECIST v1.1) and continuous safety signal tracking.

14. Observation & Follow-Up Schedule

Total Duration: Follow-up continues until disease progression, withdrawal, or estimated study completion (Dec 2027) **Onsite Visit Cadence:** Every cycle (typically 21 to 28 days) **Remote Monitoring Frequency:** Inter-cycle safety checks as clinically indicated **Checklist Review Interval:** Restaging tumor imaging every 8 weeks

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]					